

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Centre of Excellence in Artificial Intelligence

2. Subject Name: Deep Learning for Spatial Data Lab L-T-P: 0-0-3 Credits: 2

3. Pre-requisites:

4. Syllabus and reference books:

Syllabus:

- Introduction to tensor operations and deep learning libraries.
- Basics of convolutional neural networks – types of convolutional layers, their implementations and uses.
- Design of convolutional network architectures for image classification, detection and segmentation.
- Visualizing convolutional filters and corresponding activation maps, adversarial attacks.
- Deploying convolutional neural networks for transfer learning.
- Self-supervised learning for spatial data.
- Generative models for spatial data

Reference Books:

1. Francois Chollet, *Deep Learning with Python*, Manning (2017).
2. Eli Stevens, Luca Antiga and Thomas Viehmann, *Deep Learning with PyTorch*, Manning (2020)
3. Jeremy Howard and Sylvain Gugger, *Deep Learning for Coders with FastAI and PyTorch*, O'Reilly Media (2020)
4. Joshua Bengio and Aaron Courville, *Deep Learning by Ian Goodfellow*, MIT Press (2015)
5. Mohamed Elgendy, *Deep Learning for Vision Systems*, Manning (2020)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Tensor operations and gradient computation	3
2.	Overview of deep learning libraries	3
3.	Designing artificial feedforward neural networks	3
4.	Training and evaluating neural networks for classification and regression tasks.	3
5.	Introduction to convolutional neural networks	3
6.	Transfer learning with CNNs	3
7.	Visualizing CNN filters and activation maps; adversarial attacks	3
8.	CNNs for object detection	3
9.	CNNs for image segmentation and superresolution	3
10.	3D CNNs for video classification	3
11.	Autoencoders for denoting and compression	3
12.	Generative models for spatial data	3
Total number of hours		36